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Substrate cleaning device and substrate cleaning method

Abstract

A substrate cleaning device includes a substrate holder that holds a substrate, a cleaning brush that is provided to be movable between a cleaning position for cleaning the substrate held by the substrate holder and a waiting position outward of the substrate held by the substrate holder, and has a cleaning surface capable of cleaning the substrate held by the substrate holder, a remover that is configured such that liquid is transferrable to the remover, and a driver that moves at least one of the cleaning brush and the remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush.

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Background/Summary

BACKGROUND

Technical Field

(1) The present invention relates to a substrate cleaning device and a substrate cleaning method.

Description of Related Art

(2) In a substrate cleaning device, a cleaning brush is used to clean a substrate. The cleaning brush described in JP 2021-093427 A is configured to move between a cleaning position where an operation of cleaning a substrate is performed and a waiting position. In the operation of cleaning a substrate, the cleaning brush comes into contact with the substrate while a cleaning liquid is supplied to the substrate held and rotated by a substrate rotation holding device. Thus, the substrate is cleaned. In general, the cleaning brush is cleaned with a rinse liquid at the waiting position after the substrate is cleaned.

SUMMARY

(3) When the cleaning brush is moved from the waiting position to the cleaning position, droplets adhering to the cleaning brush may fall and be splashed on the surface of the substrate held by the

substrate rotation holding device. The droplets that are splashed on the surface of the substrate affect the cleanliness of the substrate.

- (4) An object of the present invention to provide a substrate cleaning device and a substrate cleaning method that enable cleaning of a substrate so as to obtain a higher level of cleanliness with a simple configuration. (1) A substrate cleaning device according to one aspect of the present invention includes a substrate holder that holds a substrate, a cleaning brush that is provided to be movable between a cleaning position for cleaning the substrate held by the substrate holder and a waiting position outward of the substrate held by the substrate holder, and has a cleaning surface capable of cleaning the substrate held by the substrate holder, a remover that is configured such that liquid is transferrable to the remover, and a driver that moves at least one of the cleaning brush and the remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush. (2) A substrate cleaning method according to another aspect of the present invention includes holding a substrate with use of a substrate holder, cleaning a substrate held by the substrate holder with use of a cleaning surface of a cleaning brush, moving at least one of the cleaning brush and a remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush.
- (5) With the present invention, it is possible to clean a substrate so as to obtain a higher level of cleanliness with a simple configuration.
- (6) Other features, elements, characteristics, and advantages of the present disclosure will become more apparent from the following description of preferred embodiments of the present disclosure with reference to the attached drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWING

- (1) FIG. 1 is a schematic side view of a substrate cleaning device according to one embodiment of the present invention;
- (2) FIG. 2 is a schematic cross sectional view for explaining a transfer operation of the substrate cleaning device;
- (3) FIG. 3 is a schematic cross sectional view for explaining the transfer operation of the substrate cleaning device;
- (4) FIG. 4 is a schematic cross sectional view for explaining the transfer operation of the substrate cleaning device;
- (5) FIG. 5 is a schematic cross sectional view for explaining the transfer operation of the substrate cleaning device;
- (6) FIG. 6 is a schematic cross sectional view mainly showing a waiting pod and a remover according to a first modified example;
- (7) FIG. 7 is a schematic cross sectional view mainly showing the configurations of a waiting pod and a remover according to a second modified example;
- (8) FIG. 8 is a schematic cross sectional view mainly showing the configurations of a waiting pod and a remover according to a third modified example;
- (9) FIG. 9 is a schematic cross sectional view mainly showing the configurations of a waiting pod and a remover according to a fourth modified example;
- (10) FIG. 10 is a schematic cross sectional view mainly showing the configurations of a waiting pod and a remover according to a fifth modified example; and

(11) FIG. 11 is a schematic cross sectional view mainly showing the configurations of a waiting pod and a remover according to a sixth modified example.

DETAILED DESCRIPTION

(12) A substrate cleaning device and a substrate cleaning method according to embodiments of the present invention will be described below in detail with reference to the drawings. In the following description, a substrate refers to a semiconductor substrate (semiconductor wafer), a substrate for an FPD (Flat Panel Display) that is used for a liquid crystal display device, an organic EL (Electro Luminescence) display device or the like, a substrate for an optical disc, a substrate for a magnetic disc, a substrate for a magneto-optical disc, a substrate for a photomask, a ceramic substrate, a substrate for a solar cell, or the like. Further, a substrate, described below, has a circular shape except for a portion in which a notch is formed in a plan view.

(1) SUBSTRATE CLEANING DEVICE

(13) FIG. 1 is a schematic plan view of the substrate cleaning device **100** according to one embodiment of the present invention. As shown in FIG. 1, the substrate cleaning device **100** includes a spin chuck **10**, a cleaning brush **20**, an arm **30**, an arm driver **40**, a brush rotation driver **50**, a waiting pod **60**, a remover **70** and a controller **80**.

(14) In the substrate cleaning device **100**, the substrate W is carried in or out by a transport robot (not shown). The spin chuck **10** includes a holder **11** and a chuck rotation driver **12**. The substrate W carried in by the transport robot is placed on the holder **11**. A plurality of suction paths (not shown) are formed in the holder **11**. When air inside the suction paths is exhausted, the lower surface of the substrate W is adsorbed under vacuum by the holder **11**. Thus, the holder **11** holds the substrate W in a horizontal attitude. The chuck rotation driver **12** is connected to the holder **11** via a connection member. The chuck rotation driver **12** includes an electric motor, for example, and is configured to be capable of rotating the holder **11**. Thus, the substrate W held by the holder **11** in a horizontal attitude is rotated.

(15) A cleaning nozzle **13** is provided above the holder **11** and supplies a cleaning liquid toward the upper surface of the substrate W held in a horizontal attitude. The cleaning liquid discharged from the cleaning nozzle **13** spreads outwardly due to a centrifugal force caused by rotation of the substrate W.

(16) The cleaning brush **20** includes a cleaning surface **21** that is opposite to the substrate W held in a horizontal attitude by the holder **11**. The cleaning brush **20** is supported by the arm **30** so as to be rotatable about an axis extending in a vertical direction. The cleaning brush **20** is supported by the arm **30** so as to be movable between a cleaning position for cleaning the substrate W held by the holder **11** and a waiting position outward of the substrate W held by the holder **11**. While the cleaning brush **20** is formed of polyvinyl alcohol (PVA), for example, the material of the cleaning brush **20** is not limited to this. Another resin material, a ceramic material or the like can be used.

(17) The arm driver **40** includes an actuator, for example, and is configured to be capable of three-dimensionally moving the cleaning brush **20** supported by the arm **30**. The brush rotation driver **50** includes an electric motor, for example, and rotates the cleaning brush **20** supported by the arm **30** about the axis extending in the vertical direction. In the present embodiment, the brush rotation driver **50** rotates the cleaning brush **20** in the direction opposite to the direction in which the substrate W held by the holder **11** is rotated. In this state, the rotating substrate W and the rotating cleaning brush **20** come into contact with each other, so that the substrate W is cleaned.

(18) At the waiting position for the cleaning brush **20**, the cylindrical waiting pod **60** is provided, for example. The cleaning brush **20** is contained in the waiting pod **60** except when the substrate W is being cleaned. A cleaning nozzle **61** is provided in the waiting pod **60**. The cleaning nozzle **61** is connected to the cleaning liquid supplier **62** and discharges a cleaning liquid to the cleaning surface **21** of the cleaning brush **20** that is contained (waiting) in the waiting pod **60**. Thus, when the cleaning brush **20** is rotated, the cleaning surface **21** is cleaned. A waste liquid hole (not shown) is formed in the bottom portion of the waiting pod **60**, and a waste liquid flow path (not shown) is

connected to the waste liquid hole. Thus, the cleaning liquid in the waiting pod **60** is drained.

(19) In the present embodiment, the remover **70** is provided to be located in the waiting pod **60**. In the present embodiment, the remover **70** is formed of a hydrophilic material (quartz glass, for example). Thus, liquid adhering to a member close to the remover **70** is likely to be transferred to the remover **70**. In the present embodiment, the remover **70** has a shape projecting upwardly and has a tip portion having a hemispherical shape. The shape of the top portion of the remover **70** is not limited to a hemispherical shape and may be another shape. The shape of the tip portion of the remover **70** may have any other shape such as a cone shape, a pyramid shape or a curved shape.

(20) Further, on the surface of the remover **70**, a plurality of slits SL that can promote the transfer of liquid from another member due to capillary action are formed by mechanical processing such as sandblasting. In the present embodiment, the plurality of slits are formed in at least the upper end surface of the remover **70**. For example, the plurality of slits may be formed to extend from the upper end surface to the side surface of the remover **70**.

(21) Further, the remover **70** is connected to an adjustment member **71**. The height of the remover **70** with respect to the bottom portion of the waiting pod **60** can be adjusted by the adjustment member **71**. In the present embodiment, the height of the remover **70** is fixed after the adjustment member **71** is moved upwardly or downwardly before the substrate cleaning device **100** is used. Thus, the height of the remover **70** is adjusted in advance by a user. Thus, the upper end surface of the remover **70** is fixed at an appropriate height after the adjustment. A specific example of application of the remover **70** will be described below.

(22) The controller **80** includes a CPU (Central Processing Unit), a RAM (Random Access Memory), a ROM (Read Only Memory), a storage device and the like. The controller **80** controls the operations of the chuck rotation driver **12**, the arm driver **40**, the brush rotation driver **50** and the cleaning liquid supplier **62**, described above.

(23) Here, the cleaning brush **20** is moved from the inside of the waiting pod **60** (waiting position) to above the substrate W in order to clean the substrate W held by the holder **11** after being cleaned in the waiting pod **60**. At this time, droplets adhering to the cleaning surface **21** of the cleaning brush **20** may fall onto the substrate W. As such, before the substrate W is moved upwardly, an operation of transferring droplets adhering to the cleaning surface **21** of the cleaning brush **20** to the remover **70** is performed. Hereinafter, an operation of transferring droplets adhering to the cleaning surface **21** to the remover **70** is referred to as a transfer operation.

(2) TRANSFER OPERATION OF SUBSTRATE CLEANING DEVICE **100**

(24) FIGS. **2** to **5** are schematic cross sectional views for explaining the transfer operation of the substrate cleaning device **100**. Note that the arm **30** is not shown in FIGS. **2** to **5**.

(25) As shown in FIG. **2**, in the waiting pod **60**, a cleaning liquid WL is first supplied from the cleaning nozzle **61** into the waiting pod **60**. In the present example, the cleaning liquid WL is discharged from the cleaning nozzle **61** into the waiting pod **60** in a horizontal direction. Thus, the cleaning liquid WL is supplied to the cleaning surface **21** of the cleaning brush **20**. The cleaning brush **20** is rotated about the axis extending in the vertical direction, so that the cleaning surface **21** of the cleaning brush **20** is cleaned with the cleaning liquid WL. When cleaning for the cleaning surface **21** is finished, rotation of the cleaning brush **20** is stopped. At this time, as shown in FIG. **3**, droplets WD of the cleaning liquid WL adhere to the cleaning surface **21** due to the surface tension of liquid.

(26) Next, as shown in FIG. **4**, after the cleaning surface **21** of the cleaning brush **20** is cleaned, the controller **80** controls the arm driver **40** to lower the cleaning brush **20** such that the cleaning surface **21** keeps a predetermined distance ΔTa from the upper end surface of the fixed remover **70**. The distance ΔTa is set larger than 0, and equal to or smaller than the thickness in an up-and-down direction of the droplets WD adhering to the cleaning surface **21**.

(27) Therefore, when the cleaning brush **20** is lowered, part of the droplets WD adhering to the cleaning surface **21** comes into contact with the upper end surface of the remover **70**. Thus, as

shown in FIG. 5, at least part of the droplets WD adhering to the cleaning surface **21** is transferred to the remover **70**. Thereafter, the cleaning brush **20** is moved to a position above the substrate W and is then lowered. Thus, the cleaning surface **21** comes into contact with the upper surface of the substrate W, and the upper surface of the substrate W rotated by the spin chuck **10** is cleaned by the rotating cleaning surface **21**.

(28) The thickness of the droplets WD in the up-and-down direction varies depending on a cleaning condition such as an area of the cleaning surface **21**, viscosity of the cleaning liquid and hydrophilicity of the cleaning surface **21**. As such, the distance ΔTa is determined by a teaching operation performed by an operator at the time of installation or maintenance of the substrate cleaning device **100**, for example. The distance ΔTa is preferably set to 0.5 mm to 1.0 mm, for example.

(3) EFFECTS OF THE EMBODIMENTS

(29) With the above-mentioned configuration, the cleaning surface **21** of the cleaning brush **20** and the remover **70** move close to each other before the substrate W is cleaned by the cleaning brush **20**. Thus, droplets adhering to the cleaning surface **21** are transferred to the remover **70** and removed from the cleaning surface **21**. Thus, when the cleaning brush **20** moves from the waiting position to the cleaning position for cleaning the upper surface of the substrate W, droplets adhering to the cleaning surface **21** of the cleaning brush **20** are prevented from falling and being splashed on the surface of the substrate W. This suppresses degradation of the cleanliness of the substrate W due to splashing droplets. As a result, the substrate cleaning device **100** can clean the substrate W so as to obtain a higher level of cleanliness with a simple configuration.

(30) Further, because the remover **70** is formed in part of the waiting pod **60**, an additional space for provision of the remover **70** is not required. Thus, miniaturization of the substrate cleaning device **100** is not prevented, and the substrate cleaning device **100** is not prevented from being made compact.

(31) Further, because the remover **70** is formed in the waiting pod **60**, droplets transferred to the remover **70** can be collected in the waiting pod **60**. Thus, it is not necessary to provide a mechanism for collecting droplets outwardly of the waiting pod **60**. As a result, complication of the configuration of the substrate cleaning device **100** is suppressed.

(32) Further, the operation of moving the cleaning brush **20** from the waiting position to the cleaning position above the substrate W and the transfer operation can be performed only by an operation of the arm driver **40**. This can suppress an increase in number of components.

(33) Further, because the transfer operation is performed with the cleaning surface **21** and the remover **70** not coming into contact with each other and maintaining the distance ΔTa from each other, generation of particles due to the contact between the cleaning surface **21** and the remover **70** is prevented. Further, transfer of contaminants such as particles from the remover **70** to the cleaning surface **21** is prevented.

(4) MODIFIED EXAMPLES

(4-1) First Modified Example

(34) While the cleaning brush **20** is moved such that the cleaning surface **21** keeps the predetermined distance ΔTa from the fixed remover **70** in the above-mentioned embodiment, the present invention is not limited to this. FIG. 6 is a schematic cross sectional view mainly showing a waiting pod **60** and a remover **70** according to a first modified example. In the first modified example, an adjustment member driver **91** controlled by a controller **80** is further included in the configuration of the substrate cleaning device **100** of FIG. 1. The adjustment member driver **91** includes a motor or the like capable of moving an adjustment member **71** upwardly and downwardly.

(35) In this case, the controller **80** controls an arm driver **40** and the adjustment member driver **91** to move both of a cleaning brush **20** and a remover **70** such that a cleaning surface **21** of the cleaning brush **20** and the remover **70** keep a predetermined distance ΔTa from each other.

Alternatively, the controller **80** controls the adjustment member driver **91** to move the remover **70** such that the remover **70** keeps the predetermined distance ΔTa from the cleaning surface **21** of the cleaning brush **20** fixed at any position.

(36) With this configuration, because the remover **70** can be moved independently of an operation of the arm driver **40**, it is possible to easily adjust the operation of the arm driver **40**.

(4-2) Second Modified Example

(37) While the remover **70** is formed integrally with the adjustment member **71** in the waiting pod **60** in the above-mentioned embodiment, the present invention is not limited to this. FIG. 7 is a schematic cross sectional view mainly showing the configuration of a waiting pod **60** and a remover **70** according to a second modified example. In the second modified example, as shown in FIG. 7, the remover **70** is fixed to the bottom portion of the waiting pod **60**, for example. In this case, because an adjustment member **71** is not required, the remover **70** can be easily formed. Further, it is possible to suppress an increase in size of the waiting pod **60** in the horizontal direction.

(4-3) Third Modified Example

(38) FIG. 8 is a schematic cross sectional view mainly showing the configuration of a waiting pod **60** and a remover **70** according to a third modified example. In the third modified example, as shown in FIG. 8, the upper end surface of the sidewall of the waiting pod **60** located in the movement path of a cleaning brush **20** from the inside (waiting position) of the waiting pod **60** to above a substrate W is used as the remover **70**. The remover **70** formed in at least part of the waiting pod **60** may be formed of a hydrophilic material or coated with a hydrophilic material.

(39) An arm driver **40** moves the cleaning brush **20** such that a cleaning surface **21** keeps a predetermined distance ΔTa from the remover **70** at the upper end surface of the waiting pod **60** when the cleaning brush **20** is moved from the inside (waiting position) of the waiting pod **60** to above a substrate W. In this case, because the upper end surface of the waiting pod **60** is used as the remover **70**, it is not necessary to provide the configuration of the remover **70** separately from the waiting pod **60**. Therefore, droplets WD adhering to the cleaning surface **21** can be transferred only with simple control.

(4-4) Fourth Modified Example

(40) FIG. 9 is a schematic cross sectional view mainly showing the configuration of a waiting pod **60** and a remover **70** according to a fourth modified example. In the fourth modified example, as shown in FIG. 9, in a substrate cleaning device **100**, a remover **70** formed of a bar-shaped member is provided in the movement path of a cleaning brush **20** from the inside (waiting position) of the waiting pod **60** to above a substrate W. In the present example, the remover **70** is arranged so as to intersect with the movement path of the cleaning brush **20**.

(41) In this case, an arm driver **40** moves the cleaning brush **20** such that a cleaning surface **21** keeps a predetermined distance ΔTa from the remover **70** when the cleaning brush **20** is moved from the inside (waiting position) of the waiting pod **60** to above a substrate W. Thus, it is possible to transfer droplets WD adhering to the cleaning surface **21** when the cleaning brush **20** is moved from the inside of the waiting pod **60** (waiting position) to the position for cleaning the substrate W. Therefore, it is possible to transfer the droplets WD adhering to the cleaning surface **21** with simple control. The remover may be configured to be movable to keep the predetermined distance ΔTa from the cleaning surface **21** with the cleaning brush **20** being stationary at a predetermined position.

(4-5) Fifth Modified Example

(42) FIG. 10 is a schematic cross sectional view mainly showing a waiting pod **60** and a remover **70** according to a fifth modified example. In FIG. 10, the vertical cross section of the remover **70** is shown. In the fifth modified example, as shown in FIG. 10, a suction driver **92** controlled by a controller **80** is further provided in addition to the remover **70** of FIG. 7.

(43) Inside of the remover **70**, a flow path **72** that penetrates the bottom portion of the waiting pod

60 from the upper end surface of the remover **70** is formed. A suction driver **92** is connected to the flow path **72**. The suction driver **92** is a vacuum pump, for example. In this case, droplets **WD** adhering to a cleaning surface **21** are sucked by the suction driver **92** with a cleaning surface **21** keeping a predetermined distance ΔTa from the remover **70**.

(44) With this configuration, even in a case in which the distance ΔTa between the cleaning surface **21** and the remover **70** changes due to a change of the cleaning surface **21** over time, the droplets **WD** can be transferred from the cleaning surface **21** to the remover **70**. Further, even in a case in which the thickness in the up-and-down direction of the droplets **WD** adhering to the cleaning surface **21** is smaller than the distance ΔTa , it is possible to transfer the droplets **WD** from the cleaning surface **21** to the remover **70**. Therefore, it is possible to remove the droplets **WD** in various sizes adhering to the cleaning surface **21** from the cleaning surface **21**. The flow path **72** may be provided in the remover **70** of FIGS. **1** to **7**. In this case, the flow path **72** is connected to the suction driver **92** through the adjustment member **71**.

(4-6) Sixth Modified Example

(45) FIG. **11** is a schematic cross sectional view mainly showing the configuration of a waiting pod **60** and a remover **70** according to a sixth modified example. While the predetermined distance ΔTa is determined by a teaching operation performed by an operator in the above-mentioned embodiment, the present invention is not limited to this. In the sixth modified example, as shown in FIG. **11**, a substrate cleaning device **100** further includes a detector **93** that is controlled by a controller **80**. The detector **93** includes a transmission-type or reflection-type position sensor that detects the presence or absence of an object using laser light, for example. The controller **80** includes a calculator **81** that calculates the distance ΔTa . The calculator **81** is implemented by execution of a transfer operation program stored in a ROM or a storage device by the CPU of the controller **80**. Further, the calculator **81** may be implemented by hardware such as an electronic circuit. Separately from the above-mentioned transfer operation, the detector **93** detects a cleaning surface **21** of a cleaning brush **20** located at a predetermined height. This enables detection of a change amount ΔDT (an amount of wear, for example) of the cleaning surface **21** due to long-term use. The calculator **81** calculates the relative positional relationship between the cleaning brush **20** and the remover **70** based on the change amount ΔDT of the cleaning surface **21** such that the distance between the cleaning surface **21** and the remover **70** is maintained at the distance ΔTa . The controller **80** controls an arm driver **40** based on the positional relationship calculated by the calculator **81** during a transfer operation. The detector **93** may detect a change amount (amount of wear) of the remover **70**. In this case, the calculator **81** calculates the relative positional relationship between the cleaning brush **20** and the remover **70** based on the change amount of the remover **70** such that the distance between the cleaning surface **21** and the remover **70** is maintained at the distance ΔTa .

(46) With this configuration, even in a case in which at least one of the cleaning surface **21** and the remover **70** is worn due to long-term use, the distance between the cleaning surface **21** and the remover **70** can be maintained constant during a transfer operation. Therefore, it is possible to remove droplets from the cleaning surface **21** over a long period of time.

(5) Other Embodiments

(47) (5-1) While the cleaning brush **20** is used to clean the upper surface of the substrate **W** in the above-mentioned embodiment, the present invention is not limited to this. For example, the cleaning brush **20** may be used to clean another surface such as a bevel region of the substrate **W**.

(5-2) While a transfer operation is performed when the cleaning surface **21** and remover **70** come close to each other so as to maintain the distance ΔTa in the above-mentioned embodiment, the present invention is not limited to this. For example, in a case in which particles do not adhere to the remover **70**, or the like, a transfer operation may be performed with the cleaning surface **21** and the remover **70** being in contact with each other. (5-3) While the plurality of slits **SL** are formed so as to extend from the upper end surface to the side surface of the remover **70** in the above-

mentioned embodiment, the present invention is not limited to this. For example, in a case in which the remover **70** has a flat upper end surface, a plurality of slits SL may be formed only in the upper end surface. Further, a single slit SL may be formed in the remover **70**, or a slit SL does not have to be formed.

(6) CORRESPONDENCES BETWEEN CONSTITUENT ELEMENTS IN CLAIMS AND PARTS IN PREFERRED EMBODIMENTS

(48) In the following paragraphs, non-limiting examples of correspondences between various elements recited in the claims below and those described above with respect to various preferred embodiments of the present disclosure are explained. In the above-mentioned embodiment, the spin chuck **10** is an example of a substrate holder, the arm driver **40** and the adjustment member driver **91** are examples of a driver, the waiting pod **60** is an example of a container, and the flow path **72** and the suction driver **92** are examples of a sucker.

(7) OVERVIEW OF EMBODIMENTS

(49) (Item 1) A substrate cleaning device according to one aspect includes a substrate holder that holds a substrate, a cleaning brush that is provided to be movable between a cleaning position for cleaning the substrate held by the substrate holder and a waiting position outward of the substrate held by the substrate holder, and has a cleaning surface capable of cleaning the substrate held by the substrate holder, a remover that is configured such that liquid is transferrable to the remover, and a driver that moves at least one of the cleaning brush and the remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush.

(50) With the substrate cleaning device according to item 1, before the substrate is cleaned by the cleaning brush, the cleaning surface of the cleaning brush and the remover come close to each other. Thus, droplets adhering to the cleaning surface are transferred to the remover and removed from the cleaning surface. Thus, when the cleaning brush is moved from the waiting position to the cleaning position, droplets adhering to the cleaning surface of the cleaning brush are prevented from falling and being splashed on the surface of the substrate. This suppresses degradation of the cleanliness of the substrate due to splashing droplets. As a result, it is possible to clean the substrate so as to obtain a higher level of cleanliness with a simple configuration.

(51) (Item 2) The substrate cleaning device according to item 1, wherein the remover may be provided between the cleaning position and the waiting position, and the driver may move the cleaning brush from the waiting position to the cleaning position, and may move at least one of the cleaning brush and the remover such that the cleaning surface and the remover come close to each other, when the cleaning brush is moved from the waiting position to the cleaning position.

(52) With the substrate cleaning device according to item 2, because the cleaning surface and the remover come close to each other when the cleaning brush is moved from the waiting position to the cleaning position, no additional period of time is required for the cleaning surface and the remover to come close to each other. Therefore, it is possible to remove droplets from the cleaning surface while suppressing an increase in period of time required for cleaning.

(53) (Item 3) The substrate cleaning device according to item 1 or 2 may further include a container that contains the cleaning brush at the waiting position, wherein the remover may be formed in part of the container.

(54) With the substrate cleaning device according to item 3, because the remover is formed in part of the container, an additional space for provision of the remover is not required. Thus, miniaturization of the substrate cleaning device is not prevented, and the substrate cleaning device is not prevented from being made compact.

(55) (Item 4) The substrate cleaning device according to item 3, wherein the remover may be formed inside of the container.

(56) With the substrate cleaning device according to item 4, droplets transferred to the remover can be collected in the container. Thus, it is not necessary to provide a mechanism for collecting droplets outwardly of the container. As a result, complication of the configuration of the substrate cleaning device is suppressed.

(57) (Item 5) The substrate cleaning device according to item 4, wherein the driver may move the remover such that the cleaning surface and the remover come close to each other in the container.

(58) With the substrate cleaning device according to item 5, the driver moves the remover independently of an operation of moving the cleaning brush from the waiting position to the cleaning position performed by a movement mechanism. Thus, it is possible to easily adjust an operation performed by the driver.

(59) (Item 6) The substrate cleaning device according to item 4, wherein the driver may move the cleaning brush such that the cleaning surface and the remover come close to each other in the container, and may move the cleaning brush from the container to the cleaning position in order for the cleaning brush to clean the substrate.

(60) With the substrate cleaning device according to item 6, the driver performs both of an operation of bringing the cleaning surface of the cleaning brush close to the remover in the container and an operation of moving the cleaning brush from the container to the cleaning position. Thus, it is not necessary to provide a driver for bringing the cleaning surface and the remover close to each other in addition to the movement mechanism for moving the cleaning brush from the container to the cleaning position. This can suppress an increase in number of components. Further, this suppresses an increase in space in a case in which the driver is provided separately from the movement mechanism.

(61) (Item 7) The substrate cleaning device according to any one of items 4 to 6, wherein the remover may have a shape projecting with respect to the cleaning surface of the cleaning brush in the container from a bottom portion in the container.

(62) With the substrate cleaning device according to item 7, an increase in size of the container in the horizontal direction can be suppressed.

(63) (Item 8) The substrate cleaning device according to any one of items 1 to 7, wherein a sucker that sucks liquid from the cleaning surface when the cleaning surface and the remover come close to each other may be provided in the remover.

(64) With the substrate cleaning device according to item 8, even in a case in which the distance between the cleaning surface and the remover varies, it is possible to transfer droplets from the cleaning surface to the remover. Further, even in a case in which being smaller than the distance between the cleaning surface and the remover, droplets adhering to the cleaning surface can be transferred from the cleaning surface to the remover. Therefore, it is possible to remove the droplets in various sizes adhering to the cleaning surface from the cleaning surface.

(65) (Item 9) The substrate cleaning device according to any one of items 1 to 8, wherein the driver may move at least one of the cleaning brush and the remover such that a predetermined distance between the cleaning surface and the remover is maintained when the cleaning surface and the remover come close to each other.

(66) With the substrate cleaning device according to item 9, the cleaning surface of the cleaning brush and the remover are prevented from coming into contact with each other. Thus, generation of particles due to the contact between the cleaning surface and the remover is suppressed. Further, contaminants adhering to the remover are prevented from being transferred to the cleaning surface. Therefore, the substrate can be cleaned with a cleaner cleaning brush in a clean atmosphere.

(67) (Item 10) The substrate cleaning device according to item 9, may further include a detector that detects a position of the at least one of the cleaning brush and the remover when the cleaning surface and the remover come close to each other, and a controller that controls the driver based on a position detected by the detector such that the predetermined distance between the cleaning surface and the remover is maintained.

(68) With the substrate cleaning device according to item 10, even in a case in which at least one of the cleaning surface and the remover is worn, the predetermined distance between the cleaning surface and the remover is maintained. Thus, it is possible to remove droplets from the cleaning surface over a long period of time.

(69) (Item 11) The substrate cleaning device according to any one of items 1 to 10, wherein the remover may have a surface formed of a hydrophilic material.

(70) With the substrate cleaning device according to item 11, because the remover has hydrophilicity, it is possible to further promote the transfer of droplets from the cleaning surface to the remover.

(71) (Item 12) The substrate cleaning device according to any one of items 1 to 11, wherein the remover may have an opposing surface that is opposite to the cleaning surface when being close to the cleaning surface, and a plurality of slits may be formed in the opposing surface.

(72) With the substrate cleaning device according to item 12, it is possible to easily transfer droplets adhering to the cleaning surface of the cleaning brush because of capillary action caused by a slit formed in the remover.

(73) (Item 13) A substrate cleaning method according to another aspect includes holding a substrate with use of a substrate holder, cleaning a substrate held by the substrate holder with use of a cleaning surface of a cleaning brush, moving at least one of the cleaning brush and a remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush.

(74) With the substrate cleaning method according to item 13, before the substrate is cleaned by the cleaning brush, the cleaning surface of the cleaning brush and the remover come close to each other. Thus, droplets adhering to the cleaning surface are transferred to the remover and removed from the cleaning surface. Thus, when the cleaning brush is moved from the waiting position to the cleaning position, droplets adhering to the cleaning surface of the cleaning brush are prevented from falling and being splashed on the surface of the substrate. This suppresses degradation of the cleanliness of the substrate due to splashing droplets. As a result, it is possible to clean the substrate so as to obtain a higher level of cleanliness with a simple configuration.

(75) While preferred embodiments of the present disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing the scope and spirit of the present disclosure. The scope of the present disclosure, therefore, is to be determined solely by the following claims.

Claims

1. A substrate cleaning device comprising: a substrate holder that holds a substrate; a cleaning brush that is provided to be movable between a cleaning position for cleaning the substrate held by the substrate holder and a waiting position outward of the substrate held by the substrate holder, and has a cleaning surface capable of cleaning the substrate held by the substrate holder; a remover that is configured such that liquid is transferrable to the remover; and a driver that moves at least one of the cleaning brush and the remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush.
2. The substrate cleaning device according to claim 1, wherein the remover is provided between the cleaning position and the waiting position, and the driver moves the cleaning brush from the waiting position to the cleaning position, and moves at least one of the cleaning brush and the remover such that the cleaning surface and the remover come close to each other, when the

cleaning brush is moved from the waiting position to the cleaning position.

3. The substrate cleaning device according to claim 1, further comprising a container that contains the cleaning brush at the waiting position, wherein the remover is formed in part of the container.

4. The substrate cleaning device according to claim 3, wherein the remover is formed inside of the container.

5. The substrate cleaning device according to claim 4, wherein the driver moves the remover such that the cleaning surface and the remover come close to each other in the container.

6. The substrate cleaning device according to claim 4, wherein the driver moves the cleaning brush such that the cleaning surface and the remover come close to each other in the container, and moves the cleaning brush from the container to the cleaning position in order for the cleaning brush to clean the substrate.

7. The substrate cleaning device according to claim 4, wherein the remover has a shape projecting with respect to the cleaning surface of the cleaning brush in the container from a bottom portion in the container.

8. The substrate cleaning device according to claim 1, wherein a sucker that sucks liquid from the cleaning surface when the cleaning surface and the remover come close to each other is provided in the remover.

9. The substrate cleaning device according to claim 1, wherein the driver moves at least one of the cleaning brush and the remover such that a predetermined distance between the cleaning surface and the remover is maintained when the cleaning surface and the remover come close to each other.

10. The substrate cleaning device according to claim 9, further comprising: a detector that detects a position of the at least one of the cleaning brush and the remover when the cleaning surface and the remover come close to each other; and a controller that controls the driver based on a position detected by the detector such that the predetermined distance between the cleaning surface and the remover is maintained.

11. The substrate cleaning device according to claim 1, wherein the remover has a surface formed of a hydrophilic material.

12. The substrate cleaning device according to claim 1, wherein the remover has an opposing surface that is opposite to the cleaning surface when being close to the cleaning surface, and a plurality of slits are formed in the opposing surface.

13. A substrate cleaning method including: holding a substrate with use of a substrate holder; cleaning a substrate held by the substrate holder with use of a cleaning surface of a cleaning brush; moving at least one of the cleaning brush and a remover with respect to another one such that at least part of liquid adhering to the cleaning surface is transferred to the remover when the cleaning surface and the remover come close to each other at a position outward of a position above the substrate held by the substrate holder, before the substrate is cleaned by the cleaning brush.
